Freight Rate Prediction: Validation Report

Spotter ML Engineer assessment. Model: linear trend model + LightGBM on its residuals, trained on log(rate).

Summary

On the backtest fold most like the real task (train on Jan-Aug, predict Sep-Oct), the model's **mean absolute percentage error is 1.23%** (MAE $28) on loads with plausible labels, and its median bias is -0.08%. Averaged over all three 2-month folds, MAPE is 1.69%. The final model was trained on all 48,000 development loads after removing 677 corrupted labels, then used to predict all 12,000 validation loads (Nov-Dec 2025) and the 31-day December chart.

Key findings from the data

- **The validation loads come after the training period.** Development data runs Jan 1 to Oct 31, 2025, and the validation loads run Nov 1 to Dec 31, 2025. The task is a forecast, so the validation must be time-based.
- **Distance drives most of the rate**: log(rate) is about 0.87 x log(distance), so the rate per mile falls as trips get longer. Reefer costs about 13% more than Dry Van and Flatbed about 8% more.
- **Market index matters, but not fully.** The rate moves about 0.15% for each 1% move in market_index. On top of that, rates drift upward by about +7.5%/year, and they climb about 4% over the last four weeks of every quarter, then reset (Figure 1). December is a quarter-end month.
- **market_index and quote_signal are mostly date-level signals.** Each date has its own mean, plus a small amount of per-load noise. The December chart inputs do not include these two columns, so each December day's values are filled in from the validation loads on that same date.
- **Coordinates are synthetic but consistent** (one lat/lon pair per city). The validation set includes 8 cities that never appear in training (Allentown, Charlotte, Chicago, Jackson, Knoxville, Laredo, Norfolk, San Diego). The model uses coordinates instead of city names, so it can still price those lanes. In a test that removed 8 cities from training, MAPE on lanes touching them was 1.29%, against 1.23% for the other lanes.

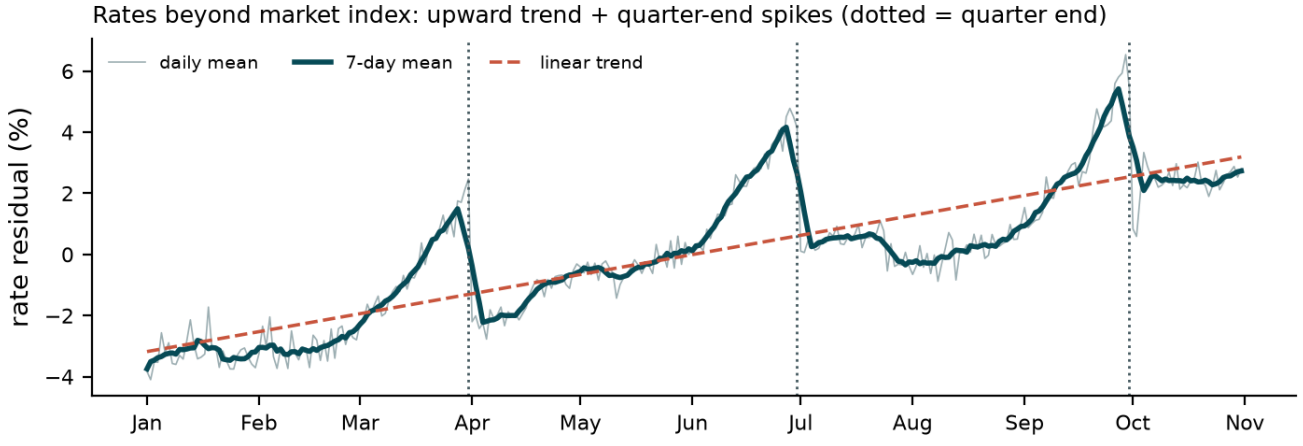


Figure 1. Daily mean log-rate residual after a regression on distance, weight, equipment and market index.

Data-quality issues and how they were handled

Issue	Development	Validation	Handling
Negative weight (sign flip)	292	145	abs(); same magnitude range as valid weights
Missing weight	300	165	equipment-type median + missing flag
Missing market_index	374	249	that day's mean market_index (date-level signal)
Corrupted posted_rate (2-5x off)	677 (1.4%)	n/a	flagged with out-of-fold model, removed from training

Corrupted labels were found by checking each load against a model that had not seen it (5-fold, Huber loss). Clean loads fall within about 5% of that model's estimate, and corrupted ones are off by a factor of 2 to 5. Nothing falls in between, so the 0.6x-1.6x cutoff does not need tuning. Removing these rows halves the backtest error (table below).

Train / validation split approach

A random split would let the model train on loads from the very days it is tested on, and it would not show how error grows the further ahead the model has to predict. So each fold copies the real setup: **train on every month before a cutoff, then test on the next two months**. The cutoffs are expanding, giving three folds (Figure 2). The label-outlier filter is re-fitted inside each fold's training window, so test labels never influence training. Metrics are reported on all test loads and on the 'clean' loads whose label is within 0.6-1.6x of the prediction. The roughly 1.4% of corrupted test labels cannot be predicted, and they dominate RMSE (about $625 for every model), which would otherwise hide real differences between models.

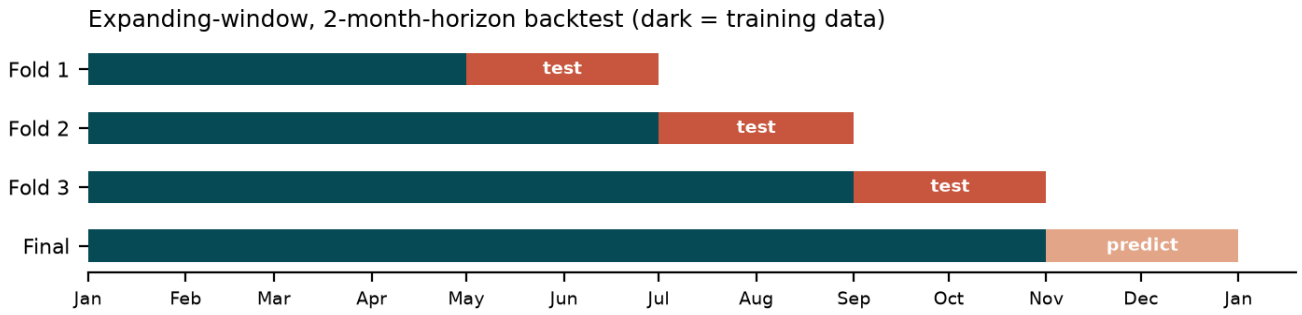


Figure 2. Backtest folds and the final train/predict split.

Backtest results: final model, per fold

Test window	Train loads	Subset	Loads	MAE ($)	RMSE ($)	MAPE	Median APE	Bias
May-Jun	19,110	all	9,696	85.0	626.1	3.57%	1.11%	+0.35%
May-Jun	19,110	clean	9,571	31.2	45.4	1.27%	1.09%	+0.35%
Jul-Aug	28,806	all	9,671	115.6	621.2	4.87%	1.99%	+1.46%
Jul-Aug	28,806	clean	9,538	64.0	101.0	2.59%	1.95%	+1.46%
Sep-Oct	38,477	all	9,523	84.3	628.2	3.77%	1.06%	-0.08%
Sep-Oct	38,477	clean	9,379	28.1	39.9	1.23%	1.05%	-0.08%

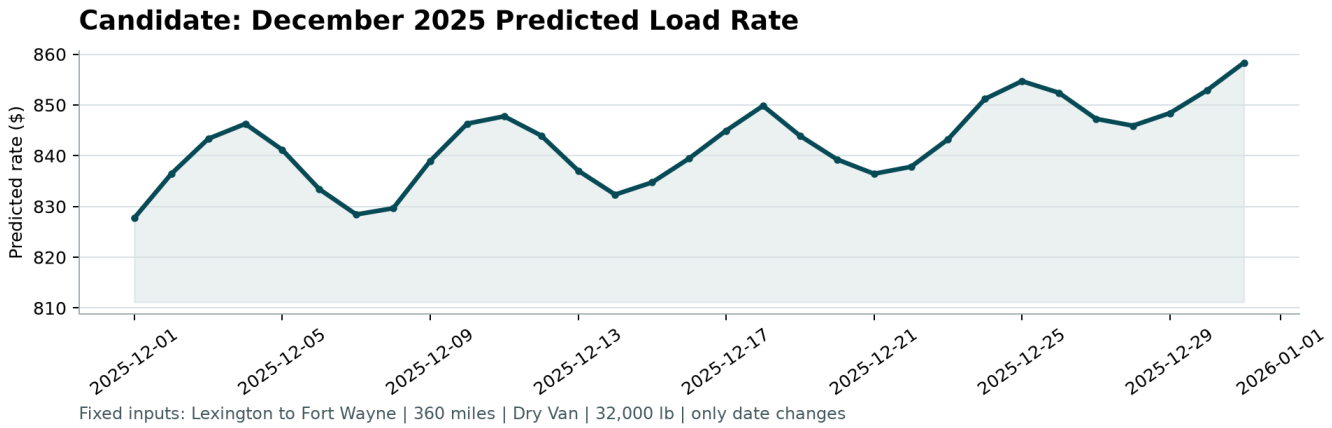
The Jul-Aug fold is the hardest. Its training window (Jan-Jun) ends right on the June quarter-end spike, and during those months market index and time rise together, so the trend slope is overestimated. With the full Jan-Oct window, the estimated slope is stable (about 0.02%/day whether training ends in Aug or Oct).

Model choice: comparison on clean loads, mean over the 3 folds

Model	MAE ($)	MAPE	Bias
final model	**41.1**	**1.69%**	**+0.58%**
linear model only	60.5	2.83%	+0.08%
GBM with date feature (no linear trend)	43.3	1.93%	-1.13%
final model without label-outlier filter	79.0	3.48%	+0.78%

Why two stages? Tree models cannot extrapolate. Past the last training date, a GBM given a date feature predicts as if it were still late October, so it falls behind the upward trend (bias -1.13%). In this model, the linear stage is the only part that sees absolute time, so it carries the trend smoothly into Nov-Dec. LightGBM then fits what is left over using lane geometry, weight, equipment, quote signal, weekday and days to quarter end. Every value those features take in Nov-Dec also appears in the training data. On its own, the linear model leaves non-linear lane effects unmodelled.

Fixed December prediction chart (produced by score.py)



Lexington to Fort Wayne, 360 miles, Dry Van, 32,000 lb. Only the date changes. City coordinates come from the data, and market_index and quote_signal are set to each date's mean across the validation loads. The weekly wave follows the weekly cycle in the December market index. The rise into late December is the quarter-end effect plus the trend.

Limitations

- Only one year of history, so yearly seasonality (for example, holiday peaks) cannot be separated from the trend. The trend is assumed to continue linearly for two more months.
- Corrupted labels in the hidden validation truth (about 1.4% if they occur at the development rate) cannot be predicted, and they will dominate RMSE.
- The December chart uses each day's average market conditions; individual loads would vary around it.